

Trekkmate Report

MAD 1 Project Report

1. Student Details

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About Me: I am a student at IIT Madras BS Degree program with an interest in web application development. I enjoy building practical applications and learning through hands-on implementation.

2. Project Details

Project Title: TrekkMate - Trekking Management System

Problem Statement:

To design and build a web-based application that allows users to browse, book, and manage treks. The system handles Admin, Staff, and User roles, manages trek availability, processes bookings, and provides separate dashboards for different roles.

Approach:

TrekkMate is built using Flask for the backend, SQLite for database storage, Jinja2 for dynamic HTML rendering, and Bootstrap/CSS for the frontend. Role-based access is used for Admin, Staff, and User dashboards. Admin manages treks, staff, users, and bookings; Staff manages assigned treks; and Users browse, book, cancel bookings, and manage profiles.

3. AI/LLM Declaration

AI/LLM assistance was used extensively during the initial development of the project, including code generation and suggestions for Flask routes, database schema, templates, and project structure. The project is being reviewed, tested, modified, and understood manually by the student before submission.

All final testing, debugging, integration, and project understanding will be completed by the student before the viva.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLite	Local database
Jinja2	Dynamic HTML templates
HTML5 / CSS3 / Bootstrap	Frontend structure, styling and responsive UI
Python 3	Backend programming language

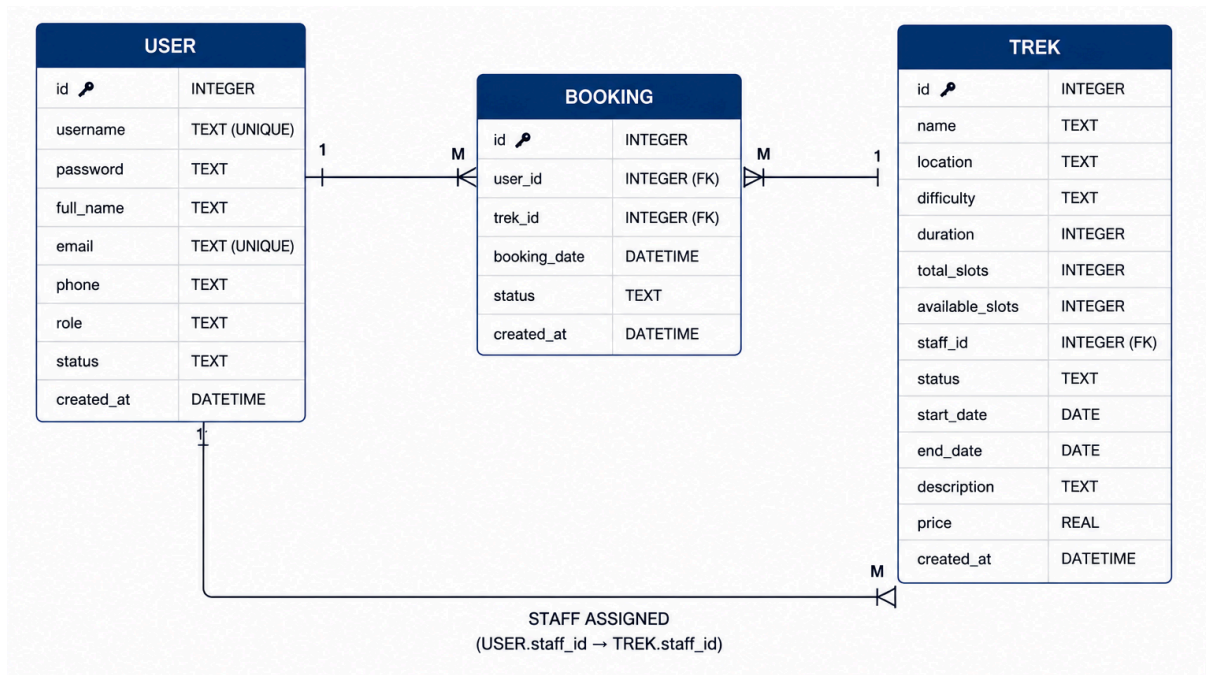
5. Database Schema / ER Diagram

Tables: User – user profile and role details; Trek – trek information, staff assignment and slot availability; Booking – user trek bookings and booking status.

Relationships: User → Booking: one user can have many bookings. Trek → Booking: one trek can have many bookings.

Staff(User) → Trek: one staff member can be assigned multiple treks.

ER DIAGRAM :



6. API Resource Endpoints

TrekkMate uses Flask application routes. The main routes relevant to the application are:

Endpoint	Method	Description
/login	GET/POST	Login
/register	GET/POST	Registration
/admin/dashboard	GET	Admin dashboard
/admin/treks	GET	Manage treks
/admin/staff	GET	Manage staff
/admin/users	GET	Manage users
/admin/bookings	GET	View bookings
/staff/dashboard	GET	Staff dashboard
/user/dashboard	GET	User dashboard
/user/treks	GET	Browse treks
/user/book/<id>	GET	Book a trek
/user/cancel/<id>	GET	Cancel booking
/user/profile	GET/POST	Edit profile

7. Architecture and Features (optional)

Architecture Overview: app.py – Flask routes and application logic; database.py – SQLite connection and table creation; seed.py – demo data; templates/ – Jinja2 pages; static/ – CSS and frontend assets.

Implemented Features: User registration/login; role-based access; admin trek/staff/user management; staff assigned-trek management; trek search/filtering; booking and cancellation; profile and booking history; dashboard statistics; flash messages; responsive UI.

8. Video Presentation

Drive Link: https://drive.google.com/file/d/1GgAsvu0yEqueuS2LzGCChg_faJ6GGS-RH/view?usp=sharing